

PRODUCT REQUIREMENTS DOCUMENT

Expertly

Peer Connect

*AI-powered structured professional matchmaking for
verified finance, tax, and legal experts across 20+
jurisdictions.*

Prepared for: CEO, Product, Engineering, UI/UX and Investor Review

Classification: Internal — Strategy & Build Specification

Version 1.0 — Draft for Review · July 2026

Table of Contents

- 1. Executive Summary
- 2. Strategic Rationale — Why Networking Features Succeed or Fail
- 3. Feature Overview & Member Journey
- 4. Time Zone & Scheduling Engine
- 5. Matching Engine
- 6. Video Meeting Infrastructure — 100ms Integration
- 7. AI Layer
- 8. Member Dashboard
- 9. Admin Dashboard
- 10. Data Model
- 11. API Specification
- 12. Wireframes
- 13. Process Flows
- 14. Analytics & KPIs
- 15. Risks, Edge Cases & Trust Safeguards
- 16. Roadmap

1. Executive Summary

Executive Explanation

Expertly today is a marketplace: a Chartered Accountant in Singapore lists availability, a client in Dubai finds them, they engage on a paid mandate. Peer Connect is a different product living inside the same platform. It is not about clients hiring experts — it is about experts meeting other experts. A tax partner in Mumbai gets matched with a transfer-pricing specialist in Frankfurt for a structured 25-minute video conversation. The platform picks the match, proposes the time, briefs both people beforehand, runs the call inside Expertly, and follows up afterwards. Nobody has to work out who to talk to, when, or what to say.

This matters commercially because professional-services marketplaces have a cold-start problem: a directory of profiles with no reason to log back in decays into an unused listing site within weeks of launch. Peer Connect gives every member — including the majority who are not actively buying or selling a mandate in any given month — a standing reason to open Expertly every week. It converts the product from something checked occasionally into something that shows up in a member's calendar, the same way a well-run mastermind group or an alumni network does, except algorithmically arranged rather than left to chance.

This document specifies Peer Connect end-to-end: the matching logic that decides who meets whom, the time-zone engine that makes a 20-jurisdiction platform schedulable without manual coordination, the in-app video experience built on 100ms, the AI layer that prepares and closes out every meeting, the member and admin dashboards, the complete data model, and the API contracts needed to build it. It also states plainly where the risk sits — no-shows, matching fairness, over-representation of legal professionals in a network we are actively trying to weight toward CA/finance/tax, and cross-border data handling — because a networking feature that is not trusted will not be used a second time.

Technical Explanation

Peer Connect is built as five cooperating services layered on the existing Expertly platform: a **Matching Service** (queue, scoring, assignment), a **Scheduling Service** (availability, time zones, calendar sync), a **Meetings Service** (100ms room lifecycle, recording, transcript ingestion), an **AI Service** (pre-meeting briefs and post-meeting summaries via LLM calls against structured context), and a **Trust & Safety layer** (no-show tracking, reporting, blocking) that reads from and writes to all of the above. All services share the existing Expertly Postgres instance under a dedicated `peer_connect` schema, so identity, membership tier, and verification status are read from the core users table rather than duplicated — this is important for a small team: one source of truth for "who is this person and are they verified."

1.1 What Is Being Built

- A matching engine that scores and pairs verified professionals across 18+ disciplines and 20+ jurisdictions on a recurring cadence (default weekly, configurable per member).
- A time-zone and availability engine that removes manual back-and-forth on scheduling, including DST handling and two-way calendar sync (Google, Outlook, Apple).
- In-app video meetings via 100ms — no external Zoom or Meet link ever leaves the platform.
- An AI layer generating a pre-meeting brief (icebreakers, shared interests, agenda, referral angles) and a post-meeting summary (action items, follow-ups, suggested next connection).
- A member-facing dashboard (networking score, countries connected, meeting history, referral hub) and an internal admin dashboard (analytics, manual override, moderation, feature flags).
- A complete relational data model and REST API surface for all of the above.

1.2 What This Is Not

- Not a replacement for the existing paid-mandate marketplace — Peer Connect is unpaid, peer-to-peer, and explicitly non-transactional. No invoicing or escrow logic is in scope for v1.
- Not a mass video-conferencing product — 100ms is used narrowly for scheduled 1:1 professional meetings, not open rooms, webinars, or broadcast.
- Not a generic "networking event" calendar — every meeting is a single algorithmically matched pair, not a group session. Group formats are a Phase 3+ consideration (see Roadmap).

1.3 Why Now

Expertly is pre-launch with no existing social presence and a professional database of 500-2,000 contacts of mixed provenance. Peer Connect is designed to be the retention mechanism a Phase-0, no-paid-budget launch cannot otherwise afford: it replaces paid re-engagement marketing with a built-in habit loop, and gives the platform a defensible reason for a Chartered Accountant to keep an Expertly membership even in months with no active client mandate.

1.4 Success Definition

Metric	Definition	Target (Month 6 post-launch)
--------	------------	------------------------------

Meeting Acceptance Rate	% of proposed matches confirmed by both parties	≥ 65%
Meeting Completion Rate	% of confirmed meetings occurring without a no-show	≥ 85%
Weekly Active Networkers	Members completing ≥1 Peer Connect meeting per week	≥ 30% of active members
Referral Conversion	% of meetings producing a logged referral or introduction	≥ 15%
Cross-Border Meeting Share	% of meetings between members in different countries	≥ 60%

2. Strategic Rationale — Why Networking Features Succeed or Fail

Executive Explanation

Most "networking" features inside professional platforms fail for one of four reasons, and it is worth naming them plainly before proposing another one.

2.1 Why Communities Fail

- **No forcing function.** A directory or a Slack group requires the member to do the work of finding someone and initiating contact. Most members never do this more than once — not because they dislike networking, but because the activation energy of "who do I message, and what do I say" is higher than the perceived payoff of a single conversation.
- **Flat, undifferentiated membership.** Communities that treat every member identically — same content, same events, same prompts — generate a wall of noise that senior professionals (the exact people Expertly needs most) opt out of within weeks. A managing partner has no patience for a general-purpose forum.
- **No accountability loop.** Without a scheduled, mutually-agreed meeting, "networking" degenerates into unanswered LinkedIn messages. There is no social cost to ignoring an open-ended prompt to "connect with someone new."
- **Value decays after the first use.** Many networking products are interesting once — you browse the directory, maybe message a couple of people — and then there is no reason to return, because nothing new is being generated for you.

2.2 Why Networking Communities Succeed

The networking products that do sustain engagement — YPO forums, structured mastermind groups, curated angel-investor dinners — share a common mechanic: someone else decides who you talk to, on a schedule, with enough context that the conversation is worth having. The organizing party (a forum chair, a curator) does three things members will not reliably do themselves: they select a compatible peer, they set a specific time, and they frame why this particular pairing is worth 25 minutes. Peer Connect is designed to be that organizing party, automated.

2.3 How Expertly Becomes Different

Expertly already has the one ingredient most networking products lack: a verified, credentialed population segmented by discipline, jurisdiction, and experience tier (Budding vs. Seasoned). This is precisely the metadata a matching algorithm needs. A generic professional network (LinkedIn) has breadth but no verification and no structured matching; an in-person forum (YPO) has structure but is geographically bound and expensive to scale; a webinar platform has reach but no 1:1 depth. Peer Connect combines verified identity, structured metadata, and automated 1:1 scheduling — a combination none of the adjacent categories offer together.

2.4 Why Video Calling Alone Is Not Enough

Handing two strangers a video link with no other support reproduces the "cold outreach" problem Expertly's own marketing explicitly positions itself against. The awkward first ninety seconds of an unstructured call — "so, tell me about yourself" — is exactly the friction that causes people to decline the second invitation. The video layer is necessary infrastructure, not the product; the product is what happens around the call: the reason the match was made, the brief that removes the cold start, and the summary that turns a conversation into a followable outcome.

2.5 Why AI-Powered Networking Creates Competitive Advantage

AI is applied at exactly the two moments where a human curator would add value and a directory cannot: before the meeting (translating structured profile data into a specific, personalized reason to talk and good opening questions) and after the meeting (turning a conversation nobody wrote notes on into action items and a next-step suggestion). This is a genuine data moat: every completed meeting produces feedback, ratings, and outcome data that improves the matching model, which competitors starting from zero cannot replicate quickly.

Technical Explanation

Concretely, the four failure modes above map to four design commitments enforced elsewhere in this document: (1) matching is push-based and scheduled server-side, not member-initiated search, removing the "who do I message" cost; (2) scoring is tier- and discipline-aware so a Seasoned M&A partner is not matched against a Budding compliance associate by default (see §5.3 weighting); (3) every match becomes a calendar-committed meeting object with reminders and a no-show consequence, not an open-ended suggestion (see §4.7, §5.6); (4) the AI Service persists a brief and a summary against every meeting, so a member's tenth meeting is qualitatively better-supported than their first, and the corpus of past meeting outcomes feeds back into the compatibility model (see §7.7).

2.6 How This Feature Improves Retention

Retention in a membership product is a function of "reasons to open the app between transactions." Peer Connect manufactures a weekly reason independent of whether a member has an active client engagement, converting Expertly from a transactional tool into a habitual one. A member who has had five good Peer Connect conversations has five reasons not to let their annual membership lapse, distinct from whatever mandate work brought them to the platform originally.

2.7 How This Feature Increases Referrals

Every structured 1:1 between two verified professionals in adjacent or complementary disciplines is a referral opportunity by construction — a tax advisor talking to an M&A lawyer, a fractional CFO talking to a compliance specialist. The AI brief explicitly surfaces "referral opportunities" as a pre-meeting agenda item (§7.2), and the Referral Hub (§8.7) gives members a low-friction way to log and track an introduction that came out of a call, which in turn feeds the referral-conversion metric tracked in §14.

2.8 How This Feature Builds Trust

Trust here has two layers. First, platform trust: every participant is a verified member of the same credentialed network the client-facing marketplace already vets, so a meeting request from Expertly is categorically different from a cold LinkedIn message. Second, interpersonal trust: the pre-meeting brief and shared-context framing mean each side arrives already knowing why the conversation is worth having, which shortens the distance from stranger to professional acquaintance and increases the odds either side would vouch for the other later.

2.9 How This Feature Creates Network Effects

Each new member joining a jurisdiction or discipline increases match quality — and therefore perceived value — for every existing member who could plausibly be matched with them. A platform with three tax specialists in the UAE has a thin match pool; a platform with thirty has real optionality on time zone, seniority, and sub-specialty. Because the compatibility model also improves with completed-meeting outcome data (§5.4), the matching itself gets better as volume grows — a second, data-driven layer of network effect on top of the simple supply-side one.

3. Feature Overview & Member Journey

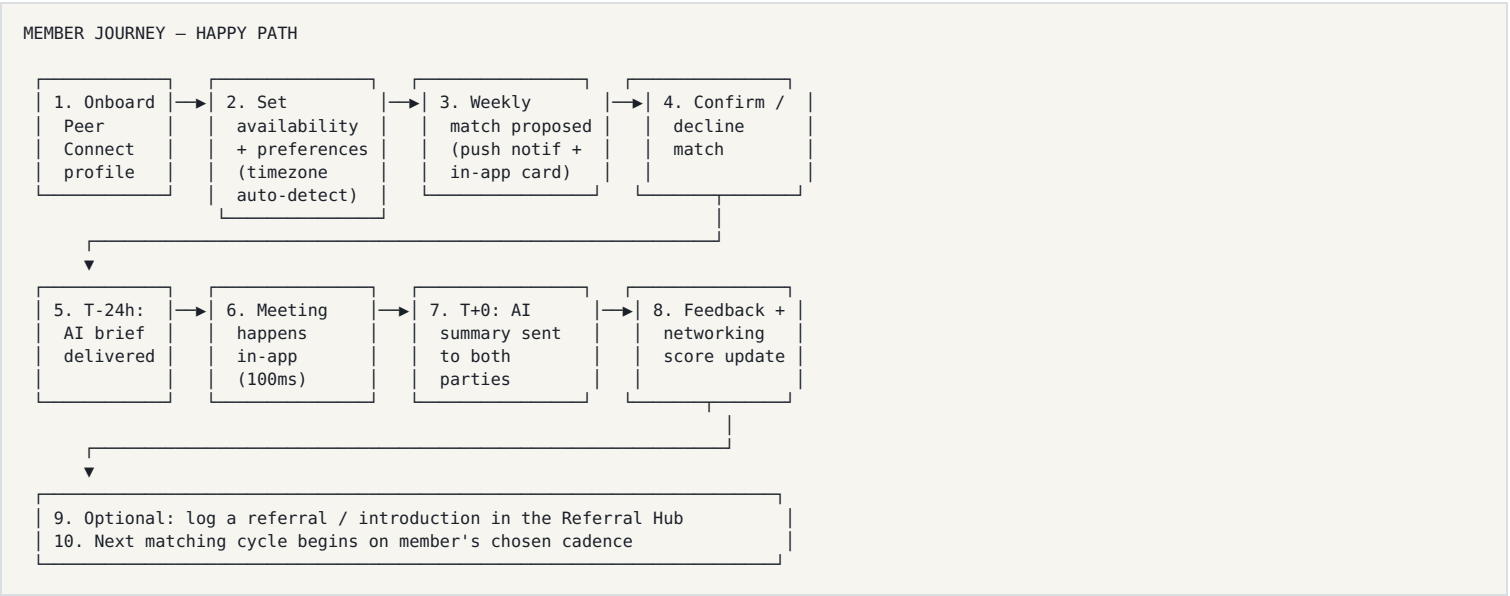
Executive Explanation

From a member's point of view, Peer Connect works like this: they set their availability once and update it occasionally. Every week (or at whatever cadence they choose), Expertly proposes a match — one specific person, one specific time slot that works for both time zones. The member confirms or declines. If confirmed, a day before the call they receive a short brief: who this is, why the platform thinks the conversation is worth having, and three good questions to open with. The call happens inside Expertly, in the browser, with no separate app to install. Afterwards, both people get a summary of what was discussed and what, if anything, they agreed to follow up on. Over time, a dashboard shows how many countries they've connected with, how many meetings they've completed, and whether any of those conversations turned into a referral.

Technical Explanation

The ten inputs to the matching score are: Availability overlap, Time zone distance, Practice area (discipline) complementarity or overlap, Industry focus, Years of experience / tier, Languages spoken, stated Networking Goals, Previous Meeting history (to avoid repeats), Peer Rating, and Country diversity contribution. These are detailed with weights and formulae in §5.3. The system is push-based: a scheduled job enqueues eligible members into the Matching Queue on each member's chosen cadence, the Matching Service scores candidate pairs, the Scheduling Service resolves a mutually available slot, and the Meetings Service provisions a 100ms room at T-0. The AI Service is invoked twice per meeting lifecycle: once at T-24h (brief generation) and once at T+0 after the call ends (summary generation), both detailed in §7.

3.1 End-to-End Member Journey



3.2 Matching Input Signals

Signal	Source	Used For
Availability	Availability Grid (§4.4)	Filter to only mutually-available time slots
Time zone	Profile IANA tz + Availability Grid	Score proximity; drives "expanding search radius" (§4.6)
Practice areas	Profile disciplines (18+)	Complementarity scoring — adjacent, not identical, disciplines score highest
Industries	Profile industry tags	Secondary relevance signal within a discipline
Experience / tier	Budding / Seasoned tier + years	Tier-aware weighting so seniority gaps are intentional, not accidental
Languages	Profile languages	Hard filter — no match without ≥1 shared working language
Networking goals	Member-selected tags (e.g. "find referral partners," "explore new markets")	Boosts pairs with complementary stated goals
Previous meetings	Matches / Meetings history table	Cooldown penalty against repeat pairing (§5.7)
Ratings	Meeting Feedback table	Down-weights members with poor reliability/rating history
Country diversity	Aggregated country-pair history	Rewards under-represented cross-border pairings (§5.5)

3.3 AI Touchpoints Across the Journey

- **Pre-meeting (T-24h):** icebreakers, conversation guide, shared interests, meeting agenda, collaboration ideas, referral opportunities (§7.2–7.6).
- **Post-meeting (T+0):** meeting summary, action items, follow-up suggestions, suggested next connection (§7.7–7.9).

3.4 Non-Functional Requirements (Platform-Wide)

Requirement	Target
Matching job latency	Full weekly matching run for 10,000 eligible members completes in < 15 minutes
Scheduling API p95 latency	< 400ms
100ms room join time	< 3 seconds from click to live video, 95th percentile
AI brief generation	Generated and stored ≥ 12 hours before meeting start; regenerated if profile data changes
Availability sync consistency	External calendar sync reflected within 5 minutes of a change (webhook-based, not polling-only)
Data residency	Meeting recordings and transcripts encrypted at rest; retained 90 days by default, configurable per jurisdiction's professional-conduct rules
Availability	99.5% uptime for Scheduling and Meetings services (video calls are time-committed; degraded matching is tolerable, degraded meetings are not)

4. Time Zone & Scheduling Engine

Executive Explanation

Expertly's members are spread across 20+ jurisdictions, which means Peer Connect has to solve a problem most scheduling tools quietly avoid: reliably finding a time that works for two people who may be nine or twelve hours apart, without either of them doing mental time-zone arithmetic, and without breaking when a country's clocks change for daylight saving. The engine's job is to make "when should we meet" a solved problem the member never has to think about — they simply say "I'm free weekday mornings" once, in their own local time, and the system handles everything else, including keeping that promise correct through DST transitions and syncing with whatever calendar they actually use.

Technical Explanation

4.1 UTC Storage

Every timestamp that represents a fixed point in time — a scheduled meeting's start time, a token expiry, a "matched at" timestamp — is stored in the database as UTC (Postgres `timestamptz`). This is non-negotiable: local-time storage is the single most common source of scheduling bugs, because a naive local timestamp becomes ambiguous the moment DST shifts or the moment it's read by a service in a different region. Recurring, member-relative concepts (a weekly availability grid, "9am-12pm every weekday") are stored differently — as a time-of-day plus an IANA zone reference, not as a fixed UTC instant — because the whole point of a recurring rule is that its UTC equivalent changes twice a year in DST-observing regions.

4.2 IANA Time Zones

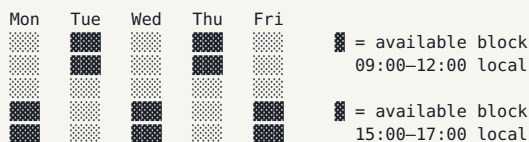
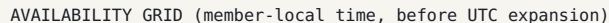
Every member profile stores a single IANA time zone identifier (e.g. Asia/Kolkata, Europe/Berlin, America/New_York) rather than a raw UTC offset. Offsets alone (e.g. "UTC+2") are insufficient because they don't encode DST rules or historical changes — Europe/London is UTC+0 in winter and UTC+1 in summer, and only an IANA identifier plus a timezone database (the system uses tzdata via each language runtime's standard library — Intl in Node, zoneinfo in Python) can resolve that correctly for an arbitrary future date. The IANA zone is auto-detected from the browser at signup and editable in Settings; a change to the profile's IANA zone re-renders (not re-stores) all displayed times, since the underlying UTC instants for already-confirmed meetings never change.

4.3 Daylight Saving Time

DST is handled at two distinct points, and conflating them is the most common implementation mistake: (1) **Availability rules** are DST-relative — "I'm free 9am-12pm on Tuesdays" means 9am in whatever the member's local wall-clock says that Tuesday, so the system must recompute the UTC equivalent per-occurrence, not once. (2) **Confirmed meetings** are DST-frozen — once a specific meeting is confirmed for a UTC instant, it does not move even if the member's zone changes clocks between confirmation and meeting day; instead, the member-facing display simply reflects whatever local time that UTC instant now corresponds to, and the system proactively flags the meeting ("this is now 1 hour earlier in your local time due to a clock change") if a DST transition falls between confirmation and meeting date.

4.4 Availability Grid

Each member defines availability as a recurring weekly grid, not individual date-specific slots — this scales to a member who wants to say "generally free Tuesday and Thursday mornings" without maintaining that manually every week. The grid is stored as a set of (day_of_week, start_local_time, end_local_time, iana_zone) rows plus optional date-specific overrides (blackout dates, one-off extra availability) in a separate table. At match time, the Scheduling Service expands the relevant grid rows into concrete UTC intervals for the target week before intersecting them against the counterpart's grid.



4.5 Country Clusters & Working Hours

To keep the matching search space tractable, jurisdictions are grouped into eight rolling clusters by UTC offset banding (e.g. GMT-5 to GMT-8 "Americas," GMT+0 to GMT+3 "Europe/Africa," GMT+4 to GMT+6 "Middle East/South Asia," GMT+7 to GMT+9 "East Asia," GMT+10 to GMT+12 "Oceania"). Default professional working hours per cluster (typically 08:00-19:00 local) bound the slots the Scheduling Service will ever propose — Peer Connect will not propose a 6am or 10pm meeting to either party, even if their stated availability technically includes it, unless the member explicitly opts into "flexible hours" in Settings.

4.6 Rare Country Strategy & Expanding Search Radius

For jurisdictions with very few verified members (a "rare country" is defined as fewer than 15 active members in a discipline cluster), a same-cluster match may not exist in a given week. The engine handles this with a progressively expanding search: it first searches within the member's own country, then within the member's regional cluster, then across clusters with at least a two-hour overlap window, and finally — if no match is found after three expansion steps — across any cluster with at least a 30-minute overlap window, explicitly flagging to the member that this is a "stretch match" on time zone. This is preferable to leaving a rare-country member unmatched, since going unmatched for successive cycles is itself a churn risk this document is trying to avoid.

EXPANDING SEARCH RADIUS

Step 1: Same country → found? yes → STOP
Step 2: Same regional cluster → found? yes → STOP
Step 3: Adjacent cluster, ≥2h overlap → found? yes → STOP
Step 4: Any cluster, ≥30min overlap → found? yes → STOP (flag: "stretch match")
Step 5: No match this cycle → member enters Priority Queue for next cycle (§5.8)

4.7 Matching Windows

Each matching cycle opens a fixed proposal window: matches are computed and proposed 5-7 days before the target meeting slot, both parties have 48 hours to confirm, and if either declines or fails to respond within the window the slot is released and both members re-enter the following cycle's queue (with a small priority boost, §5.8, so a member is not penalized for their counterpart's non-response).

4.8 Calendar Sync — Google, Outlook, Apple

Two-way calendar sync serves two purposes: reading busy/free blocks to avoid proposing a conflicting time, and writing a confirmed Peer Connect meeting into the member's primary calendar as a normal event (with the 100ms room link in the event body).

Provider	Read (busy/free)	Write (event creation)	Sync mechanism
Google Calendar	Google Calendar API freebusy.query	events.insert on primary calendar	OAuth2 + push notification channel (webhook), 5-minute fallback poll
Microsoft Outlook / 365	Microsoft Graph getSchedule	Graph /events POST	OAuth2 + Graph change notifications (webhook), 5-minute fallback poll
Apple Calendar	CalDAV free/busy report	CalDAV PUT to member's calendar collection	CalDAV has no native push; 10-minute poll interval

Calendar sync is optional at onboarding but strongly encouraged, since members who don't connect a calendar rely solely on the in-app Availability Grid, which can drift out of date with their real schedule and increase no-shows.

4.9 Scheduling Logic

```
function proposeSlot(memberA, memberB, targetWeek):
  gridA = expandAvailabilityToUTC(memberA.availabilityGrid, targetWeek)
  gridB = expandAvailabilityToUTC(memberB.availabilityGrid, targetWeek)
  busyA = fetchCalendarBusy(memberA, targetWeek) # google/outlook/apple, if connected
  busyB = fetchCalendarBusy(memberB, targetWeek)

  freeA = subtractIntervals(gridA, busyA)
  freeB = subtractIntervals(gridB, busyB)
  overlap = intersectIntervals(freeA, freeB)

  overlap = filterByWorkingHours(overlap, memberA.cluster, memberB.cluster)
  if overlap.isEmpty():
    return NO_SLOT_FOUND # triggers expanding search radius, §4.6

  slot = pickSlot(overlap, durationMinutes = 25, preference = "earliest_mutual")
  return slot # UTC instant, stored on the Meeting record
```

4.10 Rescheduling Logic & Conflict Resolution

Either party may request a reschedule up to 12 hours before a confirmed meeting. A reschedule request re-runs proposeSlot against the same matched pair (not a new match — the pairing itself is not re-scored) with the constraint that the new slot must fall within the original ± 7 day window, to avoid rescheduling silently drifting the "weekly" cadence. If a genuine conflict arises — e.g. a calendar-sync write creates a double-booking because an external event was added after the slot was confirmed — the system detects it via the next calendar poll/webhook, flags the meeting as "at risk," and proactively prompts both parties to confirm or reschedule rather than silently letting a no-show occur.

5. Matching Engine

Executive Explanation

The matching engine is the part of Peer Connect that decides, out of every other verified member on the platform, who a given person should talk to this cycle. It is not a search box — members don't browse and pick. Instead, once a week (by default), the system looks at everyone who wants a match, scores every plausible pairing against ten factors — availability, time zone, practice area, industry, experience level, language, stated goals, meeting history, ratings, and country diversity — and assigns the best available pairing to each person. It deliberately favors variety (not matching the same two people twice in a row), fairness (not always favoring the most senior or most active members), and cross-border diversity (favoring pairs across countries over pairs within the same city), because those are the properties that make the network actually feel global rather than parochial.

Technical Explanation

5.1 Matching Queue

Members opt into a cadence (weekly, biweekly, or monthly) in Settings. A scheduled job (cron, run every cycle boundary) enqueues every member whose cadence is due into the `matching_queue` table with status pending. The queue is processed in a single batch run per cycle rather than continuously, because compatibility scoring benefits from seeing the full pool at once (this is what allows fairness and diversity constraints in §5.5–5.6 to be enforced globally rather than greedily per-pair).

5.2 Compatibility Score

For any two eligible members A and B, a compatibility score is computed as a weighted sum of ten normalized (0–1) sub-scores:

```
score(A,B) =
  0.20 * availabilityOverlapScore(A,B)      # size of mutual free-time window
+ 0.12 * timezoneProximityScore(A,B)        # inverse of hour distance, post-cluster
+ 0.18 * disciplineComplementarityScore(A,B) # adjacent > identical > unrelated
+ 0.08 * industryRelevanceScore(A,B)
+ 0.10 * experienceTierScore(A,B)           # intentional, not accidental, seniority gaps
+ 0.07 * languageOverlapScore(A,B)          # 0 if no shared language -> hard filter, see below
+ 0.10 * goalComplementarityScore(A,B)      # e.g. "seeking referral partners" <-> "open to referrals"
+ 0.05 * noveltyScore(A,B)                  # penalty if matched recently, §5.7
+ 0.05 * ratingScore(A,B)                   # counterpart's historical reliability/rating
+ 0.05 * countryDiversityScore(A,B)         # bonus for under-represented country pairs, §5.5

HARD FILTERS (score forced to 0, pair excluded entirely):
- no shared working language
- either member has blocked the other
- either member is currently suspended/under review
- pair matched within the last N cycles (configurable cooldown, default 8)
```

5.3 Weighted Ranking & Tier Awareness

Discipline complementarity is scored on a curated adjacency matrix rather than raw string equality — a transfer-pricing specialist and an M&A tax advisor score higher than two people in the identical discipline, because the point of the meeting is often complementary expertise, not overlapping expertise. Experience-tier scoring is intentionally shaped so that Seasoned-to-Seasoned and Budding-to-Budding pairings score highest by default (peers tend to get more mutual value from peers), while Seasoned-to-Budding pairings are still eligible and scored moderately — useful for mentorship-style goals — provided both members have indicated openness to it via their stated networking goals.

5.4 AI-Assisted Matching & Machine Learning Improvements

The v1 compatibility score above is a transparent, auditable weighted formula — deliberately not a black-box model, because early on there isn't enough outcome data to train one reliably and a fully explainable score is easier to debug and defend to members who ask "why was I matched with this person." From roughly Month 4 onward, once a sufficient volume of completed meetings with feedback exists (target: 2,000+ rated meetings), a gradient-boosted ranking model (e.g. LightGBM) is trained to predict $P(\text{meeting rated } \geq 4/5 \text{ by both parties})$ using the same ten signals as features plus richer historical interaction features, and used to re-rank the top-N candidates the weighted formula surfaces — an ML re-ranker layered on top of an explainable base score, not a replacement for it, preserving auditability while improving precision.

5.5 Fair Matching & Country Diversity

Two fairness mechanisms are enforced at the batch level, not the pair level: (1) **Assignment fairness** — the batch solver (§5.6) is constrained so no single member can be left unmatched purely because a higher-scoring member "took" their best counterpart elsewhere in the batch; the solver optimizes total batch score, not sequential greedy assignment, specifically to avoid this. (2) **Country diversity bonus** — the platform tracks a rolling count of completed meetings per ordered country pair, and pairs involving an under-represented combination (e.g. very few Nigeria-Poland meetings to date) receive a small score bonus, nudging the network toward

broader geographic mixing rather than clustering around whichever two or three countries have the most members.

5.6 The Assignment Algorithm

Within a batch, assignment is modeled as maximum-weight matching on a graph where members are nodes and compatibility scores are edge weights (a variant of the classic assignment problem, solved via a blossom/Hungarian-style algorithm for the general non-bipartite case, since any two members regardless of grouping may be paired). Hard filters (§5.2) simply remove edges from the graph before solving.

```
function runMatchingCycle(eligibleMembers):
  graph = buildGraph(eligibleMembers)
  for (A, B) in allPairs(eligibleMembers):
    if passesHardFilters(A, B):
      graph.addEdge(A, B, weight = score(A, B))

  assignment = maxWeightMatching(graph)  # e.g. networkx max_weight_matching
                                          # guarantees globally best total score,
                                          # not greedy per-member best

  unmatched = eligibleMembers - assignment.matchedMembers
  for member in unmatched:
    applyExpandingSearchRadius(member)  # §4.6
    if still unmatched:
      addToPriorityQueue(member, boost = +1)  # §5.8

  for (A, B) in assignment.pairs:
    slot = proposeSlot(A, B, targetWeek)  # §4.9
    createMatchRecord(A, B, slot, score(A,B))
    notifyBothParties(A, B, slot)
```

5.7 No-Show Handling & Avoiding Repeat Matches

A no-show (member fails to join within 5 minutes of the scheduled start) is logged against that member's reliability score, which feeds directly into ratingScore for future pairings (§5.2) — three or more no-shows in a rolling 90-day window triggers a temporary matching suspension (member can still use the rest of Expertly, but is excluded from Peer Connect cycles until they re-confirm intent in Settings). Repeat matches are suppressed via the cooldown hard filter (§5.2): the same pair will not be re-matched by the algorithm for a configurable number of cycles (default 8), though members can always manually reconnect with a past match outside the algorithmic flow via their Connections list (§8.6).

5.8 Waiting List & Priority Queue

Members who go unmatched in a cycle (rare-country stretch case with no viable slot even at maximum search radius, or an odd-numbered pool leaving one member unpaired) are placed in a Priority Queue with an incremental score boost applied in the following cycle's matching run, so no member is systematically left out cycle after cycle. The boost decays back to baseline once a successful match is confirmed.

5.9 Worked Example

Example: Priya (Seasoned, Transfer Pricing, Mumbai, IST, goals: "find referral partners") and Lukas (Seasoned, M&A Tax, Frankfurt, CET, goals: "open to referrals, explore new markets"). Discipline complementarity is high (transfer pricing ↔ M&A tax is an adjacency-matrix strong pair). Tier match is exact (Seasoned-Seasoned). Time zone distance is 3.5 hours — well within same-cluster range. Goals are complementary by design. No shared meeting history. Result: high composite score (~0.86), proposed for the earliest mutually free 25-minute slot, e.g. Tuesday 15:00 IST / 11:30 CET.

6. Video Meeting Infrastructure — 100ms Integration

Executive Explanation

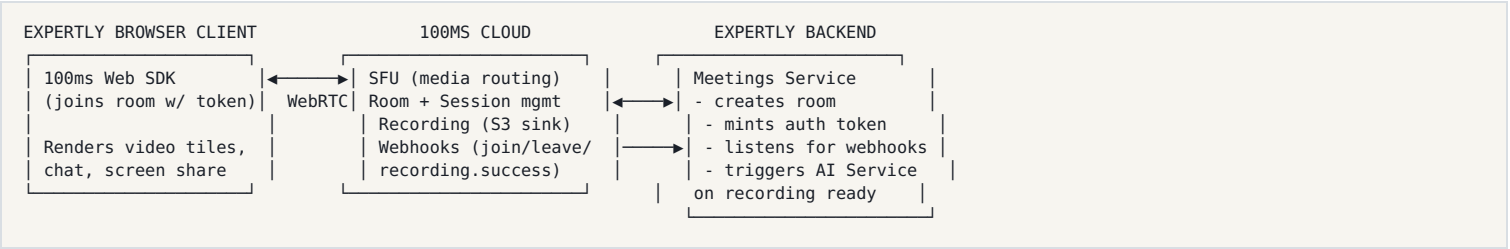
When a Peer Connect meeting starts, both members click "Join" inside Expertly and are on a live video call in their browser — no download, no external Zoom or Google Meet link, no separate login. The call is entirely inside the platform they already trust, which matters for two reasons: it keeps Expertly, not a third-party tool, as the place where the relationship happens, and it lets the platform automatically capture what it needs (a transcript for the AI summary in §7.7) without asking the member to remember to record anything themselves.

Technical Explanation

6.1 What Is 100ms and Why It Was Chosen

100ms is a managed real-time video/audio infrastructure provider (WebRTC-based SFU architecture as a service) offering client SDKs (Web, iOS, Android, React Native) plus a server-side Management API for room and token lifecycle. It was chosen over building on raw WebRTC (too much undifferentiated infrastructure work for a small team) and over embedding a third-party meeting product (see §6.10) because it is designed specifically to be embedded inside another product's UI, rather than being a standalone meeting app the member is redirected to.

6.2 Architecture



6.3 SDK & Authentication

The Web SDK (@100mslive/react-sdk) is embedded directly in the Expertly meeting-room page. Authentication is two-layered: the Expertly backend authenticates the member normally (existing session/JWT), then calls the 100ms Management API server-side to mint a short-lived **room auth token** scoped to that specific room and that specific member's role (both participants are the same role — "peer" — since there is no host/moderator distinction in a 1:1). The 100ms App Access Key and App Secret used to mint these tokens never reach the browser; only the resulting per-session token is sent to the client.

6.4 Meeting Tokens & Room Creation

```
// Server-side, at T-24h when the meeting is confirmed (not at join time)
POST https://api.100ms.live/v2/rooms
{
  "name": "peerconnect-meeting-{meetingId}",
  "description": "Expertly Peer Connect: {memberA} <> {memberB}",
  "template_id": "peer-connect-1to1-template"
}
→ 201 { "id": "room_abc123", ... }
// room_id stored on the Meeting record

// Server-side, at join time (member clicks "Join")
POST https://api.100ms.live/v2/room-codes/room/{room_id}/peer
or auth-token endpoint using App Access Key/Secret signed JWT
→ { "token": "eyJhbGciOi..." } // short TTL, single room+member scoped
// returned to the authenticated client, which passes it to the Web SDK join() call
```

6.5 Joining the Room, Screen Share & Chat

The client calls the SDK's join({ authToken }), which negotiates the WebRTC connection to the nearest 100ms SFU edge. Screen share and in-call text chat are both native 100ms SDK capabilities enabled on the room template — screen share is relevant for Peer Connect since referral or collaboration discussions often involve pulling up a document or deck; chat is retained as a lightweight fallback for sharing a link or email address without interrupting the call.

6.6 Recording & Transcript Pipeline

Recording is enabled by default (with explicit member consent captured at Peer Connect onboarding and a visible in-call recording indicator, per standard two-party consent practice) so the AI Service can generate the post-meeting summary. On call end, 100ms

uploads the recording to a configured S3 bucket and fires a `recording.success` webhook to the Expertly backend, which triggers transcript generation (via a speech-to-text pass) and hands the transcript to the AI Service (\$7.7). Members may opt out of recording per-meeting; if they do, no automated summary is generated for that meeting and both parties are notified of this at booking time.

6.7 Bandwidth & Scalability

Since every Peer Connect meeting is 1:1, bandwidth per call is modest (100ms adapts to ~500kbps-2.5Mbps per participant depending on network conditions, well within typical broadband and most mobile connections). Scalability is not a concern at the concurrency levels Peer Connect will see in year one — 100ms is provisioned for far larger group-call and live-streaming workloads, so a platform of scheduled 1:1s is a light load relative to its typical use case.

6.8 Pricing Model

100ms bills primarily on participant-minutes consumed, with recording and additional features billed separately per minute. For Peer Connect's usage pattern (short, scheduled 1:1 calls, default 25 minutes, two participants), the cost per meeting is predictable and low relative to the retention value of the meeting occurring — this is a variable cost that scales with actual engagement, not a fixed platform tax, which suits a \$0-paid-budget-team pre-launch and lets cost scale in step with usage as membership grows.

6.9 Alternatives Considered

Option	Why not chosen
Zoom (embedded/SDK)	Zoom's SDK exists but is heavier to embed for a narrow 1:1 use case, and Zoom's brand/UI conventions (waiting rooms, host controls) are built around group meetings, not a peer-to-peer product experience; using it also means routing member trust through a third-party brand inside Expertly's own UI.
Google Meet	Google Meet has no first-class embeddable SDK for building a custom in-app calling UI — it is designed to be used as a standalone Meet interface or via a generated link, which reintroduces the "external link" problem this section explicitly avoids.
Raw WebRTC (self-hosted)	Technically possible but requires building and operating TURN/STUN infrastructure, SFU media routing, and recording pipelines from scratch — a multi-month engineering investment not justified for a small team's v1.

7. AI Layer

Executive Explanation

Two AI-generated documents surround every Peer Connect meeting. Before the call, both members receive a short, personalized brief — who they're meeting, why the platform thinks it's worth 25 minutes, and a few good opening questions — so nobody starts from a blank page. After the call, both members receive a summary of what was actually discussed, any action items either side mentioned, and a suggestion for who to talk to next. This is the difference between "here's a stranger's LinkedIn profile, good luck" and a conversation that already has a running start and a clean finish.

Technical Explanation

7.1 Architecture Overview

The AI Service is a stateless service invoked at two lifecycle points, calling an LLM (Claude, via the standard Messages API) with a structured context payload assembled server-side — never with raw, unfiltered profile data, to keep prompts bounded and predictable. Every AI-generated artifact (brief or summary) is persisted in the `ai_history` table (§10.14) with the input context hash, model version, and output, both for auditability and to avoid regenerating identical output if profile data hasn't changed since the last generation.

7.2 Meeting Brief — Icebreakers & Suggested Questions

Generated at T-24h from a context payload of both members' disciplines, industries, stated goals, and (if available) a one-line summary of any prior meeting between them. The prompt is instructed to produce 3 icebreaker questions specific to the two professions in the pairing — not generic "tell me about yourself" prompts — grounded in the actual complementarity that drove the match (e.g. "Lukas, Priya's transfer-pricing work often intersects with cross-border M&A structuring — worth asking how she typically gets looped in during a deal.").

7.3 Conversation Guide & Shared Interests

A short structured guide (3-4 talking points) sequences the conversation: an opening icebreaker, a "shared interest" point drawn from overlapping industries or goals, a substantive discussion prompt tied to the discipline pairing, and a closing prompt about potential follow-up or referral. Shared interests are computed as the literal overlap between each member's tagged industries and goals before being handed to the LLM, so the AI is elaborating on real overlap, not inventing common ground.

7.4 Meeting Agenda

A lightweight, non-binding agenda (roughly: 3 minutes intro, 15 minutes core discussion using the conversation guide, 5 minutes referral/collaboration discussion, 2 minutes wrap-up and next steps) is included in the brief as a suggested structure, not a locked script — its purpose is to give both parties a shared mental model of how to use the 25 minutes well.

7.5 Collaboration Ideas & Referral Opportunities

Based on discipline complementarity and stated goals, the brief proposes 1-2 concrete collaboration or referral angles specific to the pairing (e.g. "Priya's clients doing outbound M&A into the EU may need exactly Lukas's structuring expertise — worth asking if he takes cross-referrals"). This is the AI touchpoint most directly tied to the referral-conversion metric in §14 and the retention/referral rationale in §2.7.

7.6 Prompt Architecture — Meeting Brief

```
SYSTEM: You are generating a pre-meeting brief for two verified professionals
matched by Expertly Peer Connect. Be specific to their disciplines and stated
goals; never invent shared history that isn't in the provided context; keep
tone collegial and professional, not salesy.

CONTEXT (structured JSON, assembled server-side):
{
  "memberA": { "discipline": "...", "industries": [...], "tier": "...",
               "goals": [...], "country": "..."},
  "memberB": { same shape },
  "sharedInterests": [...],           // computed overlap, not LLM-inferred
  "priorMeetingSummary": null,       // or prior AI summary if a past meeting exists
  "disciplineAdjacencyNote": "transfer pricing <> M&A tax: high complementarity"
}

OUTPUT (structured JSON):
{
  "icebreakers": [string, string, string],
  "conversationGuide": [string, string, string, string],
  "agenda": [{ "segment": string, "minutes": number }],
  "collaborationIdeas": [string],
  "referralOpportunities": [string]
```


7.7 Meeting Summary

Generated after the transcript is available (§6.6), the summary is a 4–6 sentence recap of what was actually discussed, derived from the transcript rather than from the pre-meeting brief (the brief is a prediction; the summary is a record). It explicitly avoids fabricating specifics not present in the transcript — the prompt instructs the model to omit rather than guess when the transcript is ambiguous or the audio quality degrades transcription confidence.

7.8 Action Items & Follow-Up Suggestions

Action items are extracted as a short bulleted list of anything either party committed to during the call ("send over the case study," "introduce to X colleague"), attributed to whichever party said it where the transcript makes that clear. Follow-up suggestions are lighter-touch — e.g. "consider connecting again in 6 weeks once the EU filing is complete" — and appear in both members' Connections view (§8.6).

7.9 Suggested Next Connection

Independent of the matching engine's own next cycle, the summary can surface a specific suggestion for a different Expertly member worth meeting next, if the conversation surfaced a specific need (e.g. Lukas mentions needing an Indian compliance contact — the AI Service can flag a candidate discipline+country combination for the next matching cycle to weight toward, feeding back into §5.2's goalComplementarityScore as an updated signal, not overriding the algorithmic assignment directly).

7.10 Follow-Up Email

Both the brief and the summary are also delivered as email (in addition to in-app), formatted for a professional inbox rather than a marketing template — a short, plain-text-forward email with the same structured content, since senior professionals are more likely to read and act on a plain email than open the app on a day with no active mandate work.

7.11 LLM Architecture & Model Choice

Both generation tasks run as single-turn, structured-output calls (JSON mode) against a mid-tier model chosen for cost/latency at this volume rather than a frontier model at the top of the pricing curve, since neither task requires deep multi-step reasoning — brief generation is templated personalization over structured inputs, and summary generation is extraction over a transcript, both well within a smaller model's capability. Prompts are versioned in source control, and a small labeled evaluation set (curated from early manually-reviewed briefs/summaries) is used to catch prompt regressions before shipping a prompt change.

7.12 Future AI (see also §16 Roadmap)

Planned but explicitly out of v1 scope: voice-tone/sentiment signals from the call feeding into the ML re-ranker (§5.4); an AI-suggested "warm intro" message a member can send to a past connection; and a lightweight AI assistant inside the Referral Hub that drafts the actual introduction email once a referral is logged.

8. Member Dashboard

Executive Explanation

The member dashboard is where Peer Connect lives day-to-day: what's coming up, who you've met, how your networking is going, and a place to log any referral that came out of it. It is designed to be checked in under a minute — a member should be able to see at a glance whether they have a meeting this week, what it's about, and whether anything needs their action.

8.1 Home / Upcoming Meetings

The default landing view. Shows the next confirmed meeting (counterpart name/title/country, countdown timer, one-tap access to the brief once available), any pending match awaiting confirmation, and a prompt to set availability if the member hasn't done so recently.

8.2 Networking Score

A composite, transparent score (0-100) combining meeting completion rate, average rating received, and diversity of connections (countries/disciplines), refreshed after every completed meeting. It exists to give members a sense of progress and to gently signal reliability expectations (a member who repeatedly no-shows sees their score visibly affected, which is more motivating than a hidden penalty).

8.3 Countries Connected

A simple visual (list or map) of every country the member has had a completed meeting with, incrementing as new cross-border meetings complete — this is a deliberately shareable, motivating artifact (members are likely to screenshot "connected with 14 countries" the way people share Duolingo streaks).

8.4 Meeting History

Chronological list of past meetings with status (completed / no-show / rescheduled / cancelled), linked to that meeting's summary and any logged referral.

8.5 Availability & Settings

Where the Availability Grid (§4.4) is configured, cadence is chosen (weekly/biweekly/monthly), calendar sync is connected (§4.8), notification preferences are set, and Peer Connect can be paused entirely without affecting the rest of the Expertly account.

8.6 Connections

A running list of everyone the member has completed a Peer Connect meeting with, each entry showing the meeting date, a one-line context reminder (from the AI summary), and a manual "reconnect" action that bypasses the cooldown filter (§5.2) since this is member-initiated, not algorithmic.

8.7 Referral Hub

Where a member logs a referral or introduction that resulted from a Peer Connect meeting — who it was, what it was for, and its status (introduced / in progress / closed). This is the direct data source for the referral-conversion metric in §14, and doubles as a lightweight personal CRM for the member's own tracking.

8.8 Analytics (Personal)

A simple personal view — meetings per month, average rating given/received, top disciplines met, referral count — distinct from the platform-wide admin analytics in §9.2.

9. Admin Dashboard

Executive Explanation

The admin dashboard is an internal tool for the Expertly team to keep Peer Connect healthy: seeing whether matching is actually working, catching problems (a jurisdiction with too few members, a member who keeps no-showing, a report of inappropriate conduct) early, and stepping in manually when the algorithm needs a human override.

9.1 User Management

Search/filter members by discipline, jurisdiction, tier, Peer Connect status (active/paused/suspended); view a member's full Peer Connect history, reliability score, and any reports filed against or by them.

9.2 Meeting Analytics

Platform-wide view of the metrics in §14 — acceptance rate, completion rate, average rating, meeting volume over time — segmented by discipline, tier, and cohort (e.g. members onboarded in a given month), so the team can see whether match quality is improving as the ML re-ranker (§5.4) comes online.

9.3 Country Analytics

Per-country member counts, per-country-pair meeting volume (feeding the diversity bonus in §5.5), and a live view of which jurisdictions are currently "rare countries" (§4.6) and therefore candidates for targeted outreach — directly useful for the founder-led outreach motion already in use for the broader Expertly launch.

9.4 No-Shows

A queue of recent no-show incidents, the affected member's cumulative count, and one-click access to apply or lift a matching suspension (§5.7) rather than waiting for the automatic threshold.

9.5 Reports & Moderation

Any in-app report filed against a member (inappropriate conduct, no-show pattern, safety concern) lands in a moderation queue with the relevant meeting's context (brief, and transcript/recording if consented and retained), and an action set: warn, suspend from Peer Connect, or escalate to full account review under existing Expertly trust & safety policy.

9.6 Manual Matching

An override tool letting an admin manually pair two specific members outside the algorithmic cycle — useful for high-touch cases (a VIP member, a specific strategic introduction the team wants to facilitate) — which still runs through the same Scheduling and Meetings pipeline, just skipping the automated assignment step.

9.7 Feature Flags

Standard flag infrastructure to roll out changes progressively — e.g. enabling the ML re-ranker (§5.4) for a percentage of the pool first, or testing a different default cadence for a new cohort — without a full deploy.

10. Data Model

All tables below live in a dedicated `peer_connect` Postgres schema. `users.id` is a foreign key into Expertly's existing core `users` table (identity, verification, membership tier already live there and are not duplicated). All timestamps are `timestampz` (UTC), per §4.1.

10.1 users (existing core table, referenced — not owned by Peer Connect)

Column	Type	Notes
id	uuid PK	Core Expertly identity
tier	enum(budding, seasoned)	Read by matching engine, §5.3
verified_at	timestampz	Peer Connect requires non-null

10.2 pc_profiles

Column	Type	Notes
id	uuid PK	
user_id	uuid FK → users.id	1:1
disciplines	text[]	From the 18+ discipline taxonomy
industries	text[]	
languages	text[]	ISO 639-1 codes
networking_goals	text[]	Controlled tag set, e.g. "seeking_referrals"
iana_timezone	text	e.g. "Asia/Kolkata", §4.2
country_code	char(2)	ISO 3166-1
cadence	enum(weekly, biweekly, monthly)	§5.1
status	enum(active, paused, suspended)	§5.7 suspension logic
flexible_hours_opt_in	boolean	§4.5

10.3 pc_availability

Column	Type	Notes
id	uuid PK	
profile_id	uuid FK → pc_profiles.id	
day_of_week	smallint (0-6)	Recurring rule, §4.4
start_local_time	time	Interpreted in profile.iana_timezone
end_local_time	time	
is_override	boolean	true = one-off date, false = recurring
override_date	date, nullable	Only set if is_override

10.4 pc_matches

Column	Type	Notes
id	uuid PK	
member_a_id	uuid FK → users.id	
member_b_id	uuid FK → users.id	
cycle_id	uuid	Groups matches from same batch run, §5.6
compatibility_score	numeric(4,3)	§5.2 output
score_breakdown	jsonb	Per-factor sub-scores, for auditability
status	enum(proposed, confirmed, declined, expired)	§4.7 matching window
search_radius_step	smallint	§4.6, 1-4, flags "stretch match"
created_at	timestampz	

10.5 pc_meetings

Column	Type	Notes
id	uuid PK	
match_id	uuid FK → pc_matches.id	
scheduled_start_utc	timestamptz	Frozen at confirmation, \$4.3
duration_minutes	smallint	Default 25
hundredms_room_id	text	\$6.4
recording_consent_a	boolean	\$6.6
recording_consent_b	boolean	
recording_url	text, nullable	S3 location if consented
status	enum(scheduled, completed, no_show_a, no_show_b, cancelled, rescheduled)	
actual_start_utc	timestamptz, nullable	
actual_end_utc	timestamptz, nullable	

10.6 pc_meeting_feedback

Column	Type	Notes
id	uuid PK	
meeting_id	uuid FK → pc_meetings.id	
rater_id	uuid FK → users.id	
rating	smallint (1-5)	Feeds \$5.4 ML training label
comment	text, nullable	
would_meet_again	boolean	

10.7 pc_connections

Column	Type	Notes
id	uuid PK	
member_a_id	uuid FK → users.id	
member_b_id	uuid FK → users.id	
first_meeting_id	uuid FK → pc_meetings.id	
last_meeting_id	uuid FK → pc_meetings.id	
meeting_count	int	

10.8 pc_networking_score

Column	Type	Notes
user_id	uuid PK, FK → users.id	
score	smallint (0-100)	\$8.2
completion_rate	numeric(4,3)	
avg_rating_received	numeric(3,2)	
countries_connected_count	int	
updated_at	timestamptz	

10.9 pc_notifications

Column	Type	Notes
id	uuid PK	
user_id	uuid FK → users.id	
type	enum(match_proposed, match_confirmed, brief_ready, meeting_reminder, summary_ready, no_show_warning, ...)	
payload	jsonb	
channel	enum(in_app, email, push)	
sent_at	timestamptz	

read_at	timestampz, nullable
---------	----------------------

10.10 pc_reports

Column	Type	Notes
id	uuid PK	
reporter_id	uuid FK → users.id	
reported_id	uuid FK → users.id	
meeting_id	uuid FK → pc_meetings.id, nullable	
reason	enum(no_show, conduct, safety, other)	\$9.5
details	text	
status	enum(open, reviewed, actioned, dismissed)	

10.11 pc_blocked_users

Column	Type	Notes
id	uuid PK	
blocker_id	uuid FK → users.id	
blocked_id	uuid FK → users.id	Read as hard filter, \$5.2
created_at	timestampz	

10.12 pc_no_shows

Column	Type	Notes
id	uuid PK	
meeting_id	uuid FK → pc_meetings.id	
user_id	uuid FK → users.id	Who no-showed
detected_at	timestampz	\$5.7, 5-minute join grace period

10.13 pc_referral_requests

Column	Type	Notes
id	uuid PK	
originating_meeting_id	uuid FK → pc_meetings.id	
referrer_id	uuid FK → users.id	
referred_to_id	uuid FK → users.id, nullable	Nullable if referral is to a non-member client
description	text	\$8.7
status	enum(logged, introduced, in_progress, closed_won, closed_lost)	

10.14 pc_ai_history

Column	Type	Notes
id	uuid PK	
meeting_id	uuid FK → pc_meetings.id	
type	enum(brief, summary)	\$7
model_version	text	
input_context_hash	text	Avoids redundant regeneration
output	jsonb	
generated_at	timestampz	

10.15 pc_meeting_notes

Column	Type	Notes

id	uuid PK			
meeting_id	uuid FK → pc_meetings.id			
author_id	uuid FK → users.id	Member's own private note, distinct from AI summary		
content	text			
created_at	timestampz			

11. API Specification

All endpoints are namespaced under `/api/v1/peer-connect/`, authenticated via the existing Expertly session/JWT (Authorization: Bearer <token>) unless noted, and return standard error envelopes: `{ "error": { "code": string, "message": string } }` with codes such as 401 unauthorized, 403 not_verified, 404 not_found, 409 conflict, and 422 validation_error.

11.1 Profile & Availability

Endpoint	Method	Purpose
/profile	GET	Fetch the member's Peer Connect profile (disciplines, goals, cadence, status)
/profile	PATCH	Update disciplines, industries, languages, goals, cadence, status
/availability	GET	Fetch current availability grid + overrides
/availability	PUT	Replace the recurring availability grid (request: array of {day_of_week, start_local_time, end_local_time})
/availability/overrides	POST	Add a one-off blackout or extra-availability date
/calendar/connect	POST	Initiate OAuth (google/outlook) or CalDAV (apple) connection; request: {provider}; response: {redirect_url}
/calendar/disconnect	DELETE	Revoke a connected calendar

11.2 Matching & Scheduling

Endpoint	Method	Purpose
/matches	GET	List the member's matches (proposed, confirmed, past); query params: status, from, to
/matches/{id}/confirm	POST	Accept a proposed match; response includes the resolved meeting slot
/matches/{id}/decline	POST	Decline a proposed match; request: {reason?}; releases slot per \$4.7
/matches/{id}/reschedule	POST	Request a new slot within the original ± 7 day window (\$4.10)
/meetings/{id}	GET	Fetch a specific meeting's detail, including brief/summary once generated

11.3 Meetings (100ms)

Endpoint	Method	Purpose
/meetings/{id}/join-token	POST	Server mints a scoped 100ms auth token for the authenticated member (\$6.4); errors 403 if requested outside the join window (T-5min to T+duration)
/meetings/{id}/end	POST	Signals call end from the client; triggers no-show detection + AI summary pipeline
/webhooks/100ms	POST	Inbound webhook (join, leave, recording.success) — authenticated via 100ms's signed webhook secret, not member auth

11.4 AI

Endpoint	Method	Purpose
/meetings/{id}/brief	GET	Fetch the generated pre-meeting brief; 404 if not yet generated (before T-24h)
/meetings/{id}/summary	GET	Fetch the generated post-meeting summary; 404 if recording/transcript still processing

11.5 Dashboard

Endpoint	Method	Purpose
/dashboard/summary	GET	Aggregated view: next meeting, networking score, countries connected count
/networking-score	GET	Full breakdown of the member's networking score (\$8.2)
/connections	GET	List of past connections with meeting counts (\$8.6)
/connections/{userId}/reconnect	POST	Manually request a new meeting with a past connection, bypassing cooldown
/referrals	GET / POST	List or log a referral (\$8.7); POST body: {originatingMeetingId, description, referredTold?}
/referrals/{id}	PATCH	Update referral status

11.6 Trust & Safety

Endpoint	Method	Purpose
/reports	POST	File a report against another member; request: {reportedId, meetingId?, reason, details}

11.7 Admin (requires admin role scope)

Endpoint	Method	Purpose
/admin/analytics/meetings	GET	Platform-wide meeting metrics, filterable by discipline/tier/cohort (§9.2, §14)
/admin/analytics/countries	GET	Per-country and per-country-pair metrics (§9.3)
/admin/no-shows	GET	Recent no-show queue with member cumulative counts
/admin/members/{id}/suspend	POST	Manually suspend/reinstate a member's Peer Connect access
/admin/reports	GET / PATCH	Moderation queue and status updates (§9.5)
/admin/manual-match	POST	Create a manual pairing outside the algorithmic cycle (§9.6); request: {memberAId, memberBId, targetWeek}
/admin/feature-flags	GET / PATCH	Read/update feature flag state (§9.7)

12. Wireframes

Low-fidelity structural wireframes for the core screens. These establish layout and information hierarchy only, not visual design.

12.1 Member Dashboard — Home

ExpertlyPeer Connect[Priya M.]

NEXT MEETING

Lukas Weber · M&A Tax · Frankfurt, DE
Tue 15:00 IST (11:30 CET)Starts in 2d 4h
[View Brief][Reschedule]

NETWORKING SCORE: 82COUNTRIES CONNECTED: 11

[Upcoming][History][Connections][Referral Hub]

12.2 Meeting Room (100ms)

Peer Connect Meeting🕒 12:34[End]

[Priya's video]

AGENDA
1. Intro (3 min)
2. Discussion (15 min)
3. Referral talk (5 min)
4. Wrap-up (2 min)
SHARED INTEREST:
Cross-border M&A structuring

[Lukas's video]

[Mute] [Camera] [Share] [Chat]

12.3 Availability Settings

AvailabilityTimezone: Asia/Kolkata

09-12

Mon

Tue

Wed

Thu

Fri

09-12

[]

[✓]

[]

[✓]

[]

12-15

[]

[]

[]

[]

[]

15-18

[✓]

[]

[✓]

[]

[✓]

Cadence: () Weekly () Biweekly () Monthly

Calendar sync: [Connect Google] [Outlook] [Apple]

12.4 Profile

Peer Connect Profile

Disciplines: [Transfer Pricing] [Tax Advisory] [+ Add]

Industries: [Manufacturing] [Tech] [+ Add]

Languages: [English] [Hindi] [+ Add]

Networking Goals: [Seeking referral partners] [+ Add]

Tier: Seasoned (12+ yrs)

12.5 Meeting Brief

Your meeting with Lukas Weber – Tue 15:00 IST

WHY THIS MATCH
Transfer pricing ↔ M&A tax structuring – strong complementarity

ICEBREAKERS

- Ask how he gets looped into cross-border deal structuring
- Ask about recent EU regulatory shifts affecting his clients

REFERRAL OPPORTUNITIES

- Clients doing outbound M&A into the EU

12.6 Admin — Analytics

Peer Connect – Admin Analytics

[Discipline ▼][Tier ▼]

Acceptance Rate68%

Completion Rate87%

Weekly Active312

Referral Conv.14%

Meetings per week



Rare countries this cycle:

Kenya (4), Vietnam (6)

13. Process Flows

13.1 Signup / Onboarding Flow

[Existing Expertly member] → [Opt into Peer Connect]

→ [Set disciplines, industries, languages, goals]

→ [Set availability grid + timezone (auto-detected)]

→ [Optional: connect calendar]

→ [Choose cadence] → [Eligible for next matching cycle]

13.2 Matching Flow

[Cycle boundary reached] → [Enqueue eligible members]

→ [Apply hard filters] → [Compute compatibility scores, §5.2]

→ [Max-weight batch assignment, §5.6]

→ [Unmatched? → expanding search radius, §4.6 → still unmatched? → priority queue, §5.8]

→ [Matched pairs → propose slot, §4.9] → [Create pc_matches record, status=proposed]

→ [Notify both members]

13.3 Scheduling / Confirmation Flow

[Match proposed] → [Member A confirms?] —No→ [Decline → release slot → re-enter queue]

↓Yes

[Member B confirms?] —No→ [Decline → release slot → re-enter queue]

↓Yes (within 48h window)

[Meeting created, 100ms room provisioned]

↓

[Written to both members' connected calendars, §4.8]

13.4 Meeting Flow

[T-24h: AI brief generated + delivered] → [T-15min: reminder notification]

→ [Both join within 5min grace?] —No (one party)→ [No-show logged, §5.7]

↓Yes

[Call in progress: video, screen share, chat] → [Call ends]

→ [Recording uploaded → transcript generated → AI summary generated, §7.7]

→ [Summary delivered to both parties]

13.5 Feedback Flow

[Meeting completed] → [Rating prompt sent to both parties]

→ [Rating + comment submitted] → [pc_meeting_feedback recorded]

→ [Networking score recalculated, §8.2] → [Rating feeds ML training set, §5.4]

13.6 AI Generation Flow

[Trigger: T-24h (brief) or recording.success webhook (summary)]

→ [Assemble structured context payload] → [Check input_context_hash for cache hit]

→ [Cache miss → call LLM, structured JSON output] → [Persist to pc_ai_history]

→ [Deliver via in-app + email, §7.10]

13.7 Notification Flow

[Event occurs: match_proposed / brief_ready / meeting_reminder / summary_ready / no_show_warning]

→ [pc_notifications row created] → [Fan out to configured channels: in_app, email, push]

→ [Member reads] → [read_at recorded]

13.8 Referral Flow

[Meeting completed] → [AI summary surfaces referral opportunity, §7.5, optional]

→ [Member opens Referral Hub] → [Logs referral: description, referred_to]

→ [Status updates: logged → introduced → in_progress → closed_won/lost]

→ [Feeds referral-conversion KPI, §14]

14. Analytics & KPIs

Executive Explanation

These are the numbers that tell the team whether Peer Connect is actually working — not just whether it shipped, but whether members are showing up, enjoying it, and getting real professional value out of it.

KPI	Formula	Why it matters
Engagement (Weekly Active Networkers)	Members completing ≥ 1 meeting / total eligible members, per week	Core retention-loop health signal, \$2.6
Acceptance Rate	Matches confirmed by both parties / matches proposed	Signals match-quality perception before the meeting even happens
Meeting Completion Rate	Meetings completed / meetings confirmed	Directly reflects no-show rate and reliability, \$5.7
Average Meeting Time	Mean actual duration of completed meetings	A meeting ending early may signal poor match quality; overrunning may signal high value
Referral Rate	Referrals logged / meetings completed	Direct measure of business value created, \$2.7
Countries Connected (platform)	Distinct country-pairs with ≥ 1 completed meeting	Tracks progress toward the cross-border, global-network vision
Networking Score (aggregate)	Mean member networking score, \$8.2	Composite health metric across completion, rating, diversity
Retention Lift	90-day membership renewal rate, Peer Connect users vs. non-users	The metric that ultimately justifies the feature's ROI to the business

Technical Explanation

All KPIs are computed via scheduled aggregation jobs reading from the tables in §10 (primarily `pc_matches`, `pc_meetings`, `pc_meeting_feedback`, `pc_referral_requests`), materialized into a reporting schema refreshed hourly, and surfaced in the Admin Analytics screens (§9.2, §9.3). Segmentation by discipline, tier, cohort, and country-pair is supported on every KPI to allow the team to spot localized problems (e.g. a specific discipline with a low acceptance rate) rather than only a platform-wide average.

15. Risks, Edge Cases & Trust Safeguards

Executive Explanation

No networking feature works if people don't trust it, so it's worth being explicit about what could go wrong and what stops it from going wrong.

15.1 Key Risks

Legal-professional over-representation. Legal professionals have historically over-applied relative to CA/finance/tax in Expertly's broader applicant pool. If unaddressed, Peer Connect's matching pool would skew the same way, diluting the CA/CFO/tax-specialist experience the platform is prioritizing. **Mitigation:** the discipline-complementarity weighting (\$5.3) and the country-diversity bonus (\$5.5) can both be tuned with a discipline-balance constraint in the batch solver (\$5.6), capping how much of any single discipline's pool feeds into a given cycle's matches, with the constraint parameters owned by the Admin Feature Flags panel (\$9.7) so the team can adjust it as the member mix shifts.

No-show erosion of trust. A member who is matched with someone who doesn't show up is unlikely to trust the next match. **Mitigation:** \$5.7's reliability scoring and matching-suspension threshold, plus proactive T-15min reminders (\$13.4) to reduce accidental no-shows before they become a trust problem.

Matching fairness perception. Members may suspect the algorithm favors more "active" or more senior members. **Mitigation:** the max-weight batch solver (\$5.6) explicitly optimizes total batch score rather than greedy sequential assignment, and the score breakdown stored per match (`pc_matches.score_breakdown`) allows the team to audit and, if needed, explain any specific pairing decision.

Cross-border data handling. Recordings and transcripts of professional conversations across 20+ jurisdictions raise data-residency and professional-conduct questions (e.g. some bar associations have confidentiality expectations around recorded conversations). **Mitigation:** explicit per-meeting recording consent from both parties (\$6.6), a default 90-day retention window, and jurisdiction-specific retention overrides configurable by the compliance/legal function ahead of any jurisdiction going live.

Rare-country stretch matches feeling low-quality. A member in an under-represented jurisdiction may repeatedly get "stretch match" pairings with poor time-zone overlap. **Mitigation:** \$4.6's transparent flagging of stretch matches (so expectations are set honestly) plus the country-analytics view (\$9.3) that lets the team prioritize founder-led outreach specifically toward under-represented jurisdictions.

15.2 Edge Cases

Edge case	Handling
Odd number of eligible members in a cycle	One member left unmatched by the solver enters the Priority Queue with a boost for next cycle, \$5.8
Member changes IANA timezone after a meeting is confirmed	Underlying UTC instant is unchanged; display re-renders in new local time; DST-crossing flag applied if relevant, \$4.3
Both members simultaneously request different reschedule times	First valid request wins; the second request re-runs against the already-updated slot, surfacing a conflict notice rather than silently overwriting
Member disconnects mid-call due to network issues	Not treated as a no-show if join was detected within the grace window; 100ms reconnect logic attempts to restore the session; if unresolved, member can rejoin the same room until scheduled end time
Member deletes/deactivates their core Expertly account	All future proposed/confirmed Peer Connect matches involving them are auto-cancelled and the counterpart notified; historical meeting records are retained per standard data-retention policy
AI brief generation fails or times out	Meeting proceeds regardless — the brief is an enhancement, not a hard dependency; a fallback generic (non-AI) agenda template is shown instead
Recording consent denied by one party	Call proceeds without recording; no AI summary is generated; both parties notified at booking time this will be the case

16. Roadmap

Phase	Scope
Phase 1 — Foundation	Core matching engine (weighted formula, no ML re-ranker), time-zone/availability engine, 100ms 1:1 meetings, basic AI brief + summary, member dashboard (upcoming, history, settings), admin analytics (read-only).
Phase 2 — Trust & Retention	Networking score, referral hub, connections/reconnect flow, no-show handling and suspension logic, moderation/reporting tools, calendar sync (Google + Outlook).
Phase 3 — Intelligence	ML re-ranker (\$5.4) once sufficient outcome data exists, Apple/CalDAV calendar sync, manual admin matching, discipline-balance constraints for fair representation (\$15.1), richer personal analytics.
Phase 4 — Scale	Feature-flagged progressive rollout tooling maturity, expanded country-cluster tuning as membership grows, deeper segmentation in admin analytics, sentiment/tone signals feeding the ranking model (\$7.12).

Future AI

AI-drafted warm-introduction messages for the Referral Hub; sentiment-aware re-ranking incorporating call-quality signals, not just post-hoc ratings; a longitudinal "relationship health" signal across repeated reconnects.

Referral Marketplace

A structured, trackable referral-fee-sharing layer between members who choose to formalize introductions made through Peer Connect — explicitly out of scope for v1 (\$1.2) but a natural extension once the Referral Hub's logging data shows sustained volume.

Enterprise Networking

Firm-level Peer Connect cohorts — e.g. a mid-size advisory firm opting its whole team into structured cross-firm networking with peer firms — as a potential enterprise/B2B extension once the individual-member product is proven.